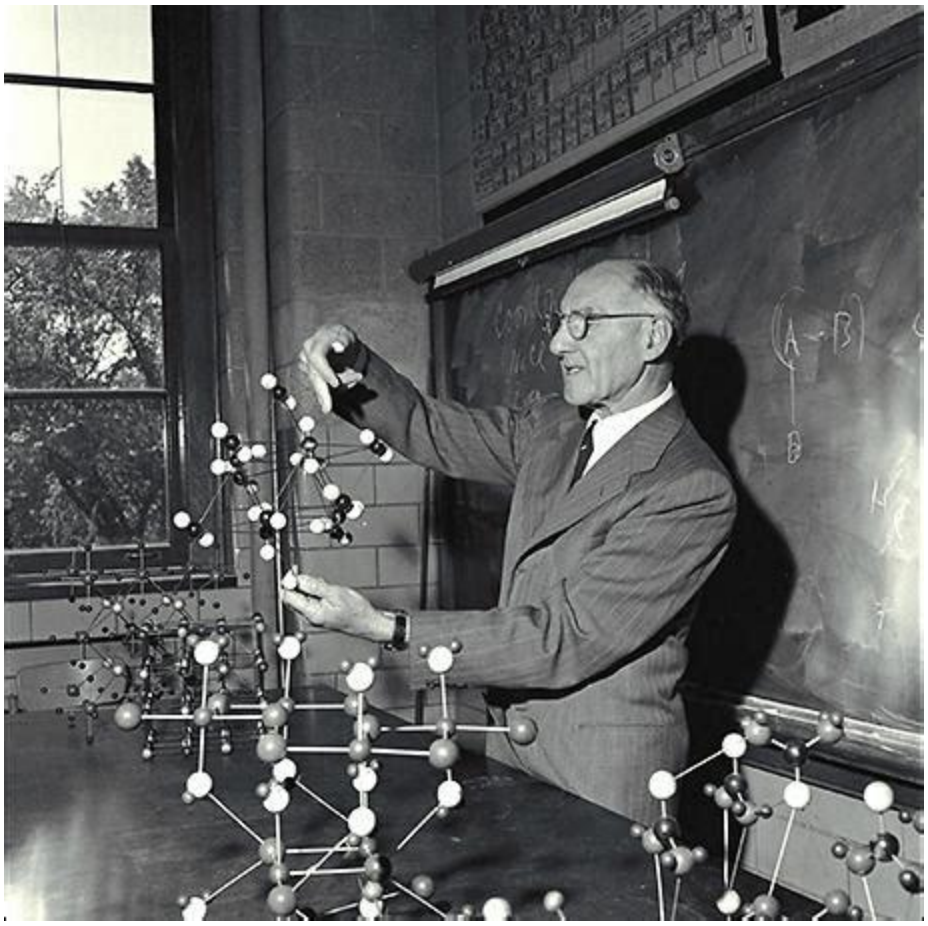

(Research materials only, per your class prompt constraints .)

A) ANCHOR SCENES (3–6)

SCENE #1: “Radioactivity bootcamp” in Rutherford’s Manchester lab

- **Summary:** Early-career pivot into radioactivity while working in Ernest Rutherford’s laboratory (a formative immersion in radioactive decay series and experimental style).
- **Date/Place:** 1910–1911 (year stated) — Manchester, UK.
- **People:** Kasimir (Kazimierz) Fajans; Ernest Rutherford.
- **Concrete details (quote-level if available):** (No direct quote from Fajans located in an open primary source about this year; treat lab anecdotes as **unverified** unless you find letters/diaries.)
- **Why it matters:** Helps explain how he quickly produced major radiochemistry results soon after (branching decay series; displacement law; isotope/element-91 work). (College of LSA)
- **Evidence quality (primary/secondary; strength):** Secondary scholarly source (history-of-science preprint) explicitly states he spent 1910–1911 with Rutherford.

- **Myth risk (low/medium/high + why): Medium** — “worked with Rutherford” is well-attested, but vivid stories about specific conversations/bench moments are often repeated without documentation.
- **Sources:**

SCENE #2: Writing down the “displacement” logic (alpha vs beta) in his 1913 paper

- **Summary:** Fajans publishes a framework connecting radioactive transformation type (α, β) to systematic shifts in periodic-table position (the core idea behind the Fajans–Soddy displacement law).
- **Date/Place:** 1913 — published in *Berichte der Deutschen Chemischen Gesellschaft* (German Chemical Society reports). (ChemTeam)
- **People:** Kasimir Fajans; (parallel/priority context: Frederick Soddy). (Wikipedia)
- **Concrete details (quote-level):** In the translated excerpt of his 1913 article, he lays out the two-class scheme (α expels a helium atom; β emits an electron) and treats it as a usable organizing principle. (ChemTeam)
- **Why it matters:** This is a highly “scene-able” moment because the paper itself contains the reasoning in his own words (a primary text you can quote). (ChemTeam)
- **Evidence quality: Primary** (Fajans’s published writing, available as a translated excerpt online). (ChemTeam)
- **Myth risk: Low** for what’s in the text; **Medium** for any claims about “the moment he realized it” beyond the paper.
- **Sources:** (ChemTeam)

SCENE #3: “Brevium” — catching a short-lived isotope of element 91

- **Summary:** Fajans and Otto/Oswald H. Göhring identify a short-lived isotope associated with element 91 in 1913 and propose the name **brevium** (emphasizing its short half-life).
- **Date/Place:** 1913 — Karlsruhe (institutional association appears in multiple summaries). (vanderkrogt.net)
- **People:** Kasimir Fajans; Otto (Oswald) H. Göhring. (vanderkrogt.net)
- **Concrete details:** The isotope is described as extremely short-lived; the “brevium” name is explicitly tied to “brief” half-life in element-history summaries. (vanderkrogt.net)
- **Why it matters:** Great backbone scene for discovery + later naming/credit complications (because later work by Hahn/Meitner and others is also tied to element 91). (math.unt.edu)
- **Evidence quality:** Mostly **secondary reference works** and element-history writeups (good for scaffolding; you’d ideally add the 1913 original

Naturwissenschaften/Physikalische Zeitschrift paper for maximum rigor). (vanderkrogt.net)
- **Myth risk: Medium** — many retellings flatten "discovery of protactinium" into a single event; the literature distinguishes isotopes and later confirmation. (math.unt.edu)
- **Sources:** (vanderkrogt.net)

SCENE #4: Leaving Munich as Germany becomes "serious," then rebuilding in Michigan

- **Summary:** After serving as director in Munich, he leaves Germany when conditions deteriorate and joins the University of Michigan, where he becomes known as an inspiring graduate teacher and prolific author.
- **Date/Place:** Munich directorship 1932–1935; departure cited as **1936** by Michigan; other summaries say he fled in **1935**. (College of LSA)
- **People:** Fajans; University of Michigan colleagues/students; (context: escalating Nazi persecution mentioned in multiple bios). (College of LSA)
- **Concrete details:** Michigan's department history explicitly says he left "in 1936, when the situation became serious in Germany," and describes him as a "brilliant and inspiring teacher… particularly at the graduate level." (College of LSA)
- **Why it matters:** Provides a human turning point (politics/institutional pressure → migration) that can frame the later "American chapter" of his science and mentorship. (College of LSA)
- **Evidence quality:** Strong **institutional secondary** source for timing/role and teaching reputation. (College of LSA)
- **Myth risk: Medium** — the exact "fled in 1935 vs left in 1936" date varies by source; treat as a discrepancy to flag rather than oversimplify. (College of LSA)
- **Sources:** (College of LSA)

SCENE #5: Late-career "quanticule theory" — embraced in industry, dismissed in academia

- **Summary:** Fajans pushes his "quanticule" approach to chemical bonding; industry invites him for seminars, while academic reception is described as hostile/indifferent, with later historians quoting Hurwic's harsh summary.
- **Date/Place:** Examples: Shell Development Labs seminars (1953); 3M visits (1952, 1956) and a week of seminars (Oct 1959); Michigan retirement context.
- **People:** Fajans; industrial research audiences; historians Hurwic (quoted) and Bulletin authors; Linus Pauling appears in a related anecdote about Fajans's attitude toward quantum mechanics study.

- **Concrete details:** The *Bulletin for the History of Chemistry* issue documents multiple industrial seminar engagements and reports that "the academic community was largely hostile…," quoting Hurwic that it was seen as "a great scientist's lapse."
- **Why it matters:** A strong conflict/controversy arc with specific, dated events (seminars) plus explicit contemporary/historical judgment.
- **Evidence quality: Scholarly secondary** (peer society history bulletin) with detailed claims and a direct quotation attribution to Hurwic.
- **Myth risk: Low–Medium** — the events are specific; the "why" behind rejection is interpretive, but the documented reception summary is source-anchored.
- **Sources:**

B) DIRECT QUOTES (3–7) — from Fajans

1. **"The starting point for these considerations is the connection that I have established between the type of a radioactive transformation and the electrochemical character of the radio element being considered."**
    - **Context/date:** 1913 paper (*Radioactive Transformations and the Periodic System of the Elements*), translated excerpt. (ChemTeam)
    - **Source (full citation):** Kasimir Fajans (1913), excerpted/translated at ChemTeam (from *Berichte der Deutschen Chemischen Gesellschaft*, Vol. 46, pp. 422–439). (ChemTeam)
2. **"It should be emphasized here that it is possible to divide all radioactive transformations into two classes…"** (followed by his α vs β description).
    - **Context/date:** 1913 paper, translated excerpt. (ChemTeam)
    - **Source:** Same as above. (ChemTeam)
3. **"…it would be meaningless to obtain a separation."** (about elements occupying the same "place" and being inseparable by chemical methods).
    - **Context/date:** 1913 paper, translated excerpt discussing chemically inseparable radioelements in the same periodic position. (ChemTeam)
    - **Source:** Same as above. (ChemTeam)

C) HUMAN REALITY DETAILS (8–15 bullets)

- He was born in Warsaw on **May 27, 1887** (as stated by the University of Michigan Chemistry department history page). (College of LSA)
- His higher education spanned Leipzig, Heidelberg, Zürich (Institute of Technology), and Manchester (per Michigan's department history). (College of LSA)
- He served as **Director of the Laboratory for Physical Chemistry at the University of Munich (1932–1935)** (Michigan page). (College of LSA)

- Michigan reports he left Munich "**in 1936**" because "the situation became serious in Germany" (institutional phrasing worth quoting carefully). (College of LSA)
- A museum/archive description notes he "**fled Germany**" and relocated to Ann Arbor, Michigan (but gives **1935**), creating a date discrepancy to flag. (digital.sciencehistory.org)
- Michigan calls him a "**brilliant and inspiring teacher… particularly at the graduate level**," and says he could "guide and inspire" student research. (College of LSA)
- Michigan credits him with publishing **nearly 200 scientific articles** and highlights how much he accomplished very early (branching series; displacement law; element-91 isotope) — useful for "compressed early brilliance" framing. (College of LSA)
- A Science History Institute digital record preserves a **postcard from Fajans to Georg Bredig**, placing him in networks of mentors/colleagues and showing personal correspondence survives in archives. (digital.sciencehistory.org)
- A second Science History Institute item is literally a postcard/photo of **Fajans with Bror Holmberg** in front of shelves of glassware — concrete lab "texture."
- A history-of-chemistry bulletin reports that, in contrast to his kindness toward students, he could be "bluntly critical" with professional peers (a personality contrast you can use carefully if you keep the wording tethered to the source).
- The same bulletin documents substantial **industry engagement** (Shell, 3M seminars), suggesting he actively sought audiences beyond academia late in life.
- The bulletin also reports a specific anecdote: he told Pauling in **1926** he "could not take the time to study" quantum mechanics (use cautiously, but it's a concrete claim in a scholarly history source).

D) SCIENCE CONNECTION (Intro Biochem / Enzymology & Protein Chemistry bridge)

- **Best-fit topic: Methods & measurement** — isotopes/radioisotopes as tracers; plus analytical chemistry methods used in biochemical labs.
- **Precise contribution (1 sentence):** Fajans helped formalize how radioactive decay changes element identity (the displacement law logic) and co-identified a short-lived isotope tied to element 91, work that sits inside the early development of isotope/radiochemistry. (ChemTeam)
- **Bridge explanation (2–4 factual sentences):** Isotopes became foundational tools for biochemistry because labeling (radioactive or stable) lets you **track atoms through reactions**—the basic idea behind tracer experiments in metabolism and enzymology. The early 20th-century radiochemistry community (including Fajans) built the conceptual and practical basis for identifying and handling isotopes (what changes in decay series; how "new" radioelements relate to the periodic table). (ChemTeam)
- **Scientist's own framing/analogy (if found):** In 1913 he frames the whole project as linking transformation "type" to "electrochemical character" — a nice "classification

mindset" quote to connect to how biochemists classify reaction mechanisms. (ChemTeam)
- **Sources:** (ChemTeam)

Note: Your roster tag says "Enzymology & Protein Chemistry," but the best-supported record positions Fajans primarily as a **physical chemist/radiochemist**; the biochem connection is strongest through **isotope methods**, not direct enzyme/protein discoveries. (College of LSA)

E) COMPLICATIONS OR CONTROVERSIES (if evidence exists)

- **Item: Priority/credit around the displacement law (Fajans vs Soddy)**
  - **What's agreed:** Both Fajans and Soddy independently arrived at a displacement-law framework around 1913; the law is commonly named for both. (Wikipedia)
  - **What's disputed/complicated:** Priority narratives vary; popular accounts can oversimplify who "did it first," and the terminology/periodic-table interpretation evolved in the literature. (Chemistry World)
  - **Sources:** (Chemistry World)
- **Item: "Discovery of protactinium" vs "discovery of an isotope of element 91"**
  - **What's agreed:** Fajans & Göhring identified an early isotope of element 91 (called "brevium" in 1913 in many summaries). (vanderkrogt.net)
  - **What's disputed/complicated:** Later work (often credited to Hahn/Meitner and others) is central to the element's broader discovery/characterization story; one-line "he discovered protactinium" claims can erase isotope and confirmation details. (math.unt.edu)
  - **Sources:** (vanderkrogt.net)
- **Item: Late-career quanticule theory reception**
  - **What's agreed:** A scholarly history bulletin documents strong negative/indifferent academic reception while industry audiences engaged via seminars.
  - **What's disputed:** Whether the rejection was mainly scientific merit vs sociological/institutional factors is interpretive; the bulletin explicitly treats the reasons as "complex."
  - **Sources:**

F) "NOT ON WIKIPEDIA" FINDS (5+)

- **Industrial seminar circuit:** Two weeks of seminars on quanticule theory at Shell (1953) and multiple 3M engagements including a week of seminars (Oct 1959). (Distinctive because it gives dated, scene-ready events.)
- **Reception quote attributed to Hurwic:** "The scientific community… regarded this theory as a great scientist's lapse…" (Distinctive because it's a blunt, source-attributed summary you can juxtapose with industry interest.)
- **Personality contrast:** The bulletin contrasts his "bluntly critical" peer interactions with "undiminished consideration" for students. (Distinctive human detail that changes how you read mentorship/lab culture.)
- **Pauling anecdote:** Report that Fajans told Linus Pauling in 1926 he "could not take the time to study" quantum mechanics. (Distinctive and specific; use cautiously but it's in a scholarly history source.)
- **Archival postcard evidence:** Science History Institute preserves a postcard from Fajans to Georg Bredig, with metadata on creation place (Munich) and later migration context. (Distinctive because it's a tangible primary artifact you could quote if you open the transcription/translation.) (digital.sciencehistory.org)
- **Photo-postcard lab texture:** Postcard image of Fajans with Bror Holmberg "behind… shelves of glassware." (Distinctive visual detail you can describe accurately.)

G) SOURCES CONSULTED (Full citations)

1. Kasimir Fajans (1913). *Radioactive Transformations and the Periodic System of the Elements* (excerpt/translation from *Berichte der Deutschen Chemischen Gesellschaft*, Vol. 46, pp. 422–439) as reproduced at ChemTeam. (ChemTeam)
2. University of Michigan, LSA Chemistry — "Kasimir J. Fajans (1887–1975)" (department history page). (College of LSA)
3. Science History Institute Digital Collections — "Postcard from Kasimir Fajans to Georg Bredig" (metadata + links to transcription/translation). (digital.sciencehistory.org)
4. Science History Institute Digital Collections — "Postcard of Kasimir Fajans and Bror Holmberg… shelves of glassware" (viewer/description).
5. *Bulletin for the History of Chemistry* (1990), full issue PDF (sections on quanticule theory reception; industry seminars; personality notes).
6. Royal Society Publishing (2020): R. G. Egdell, "Henry Moseley, X-ray spectroscopy and the periodic table" (notes element 91 gap and that a short-lived isotope was first identified by Fajans & Göhring in 1913). (Royal Society Publishing)
7. VanderKrogt Elements Multidict — Protactinium entry (credits 1913 identification and "brevium" naming rationale). (vanderkrogt.net)
8. *Chemistry World* feature — "Transmutations and isotopes" (context on displacement law and early isotope era). (Chemistry World)
9. H. Kragh (arXiv, 2011) "The isotope effect: Prediction, discussion, and discovery" (states Fajans spent 1910–1911 with Rutherford).
10. Your assignment prompt PDF (Deep Research Prompt: constraints + output format).

H) FLAGGED “MYTH-RISK” DETAILS (If any)

- **Claim:** “Fajans fled Germany in 1935.”
  - **Where it appears:** Science History Institute postcard metadata summary. (digital.sciencehistory.org)
  - **Why sourcing is weak/complicated:** Another reputable institutional source (University of Michigan Chemistry history) states he left in **1936**. (College of LSA)
  - **Best available evidence:** Treat as “mid-1930s” unless you locate a primary document (appointment letter, passport/visa, dated correspondence) pinning the exact move. (digital.sciencehistory.org)
- **Claim:** “Fajans discovered protactinium” (single-step story).
  - **Where it appears:** Common summaries and even high-level bios; Michigan notes he discovered the “first isotope of the 91st element.” (College of LSA)
  - **Why sourcing is weak/complicated:** The element-91 story involves isotopes and later confirmation; short-lived “brevium” identification (1913) is often separated from subsequent isolation/characterization in later years. (math.unt.edu)
  - **Best available evidence:** Be explicit: “identified a short-lived isotope associated with element 91 (‘brevium’)” and then note later work by others in the element’s naming/confirmation history. (vanderkrogt.net)

LENS RECOMMENDATION (after A–G)

(Using the rubric in your prompt .)

Score table (0–3 each + 1-sentence justification):

- **Eureka: 3** — The 1913 paper provides quotable, primary-text reasoning that can anchor a discovery narrative. (ChemTeam)
- **Failure: 1** — No single documented “setback scene” yet (beyond general migration pressures and later theory rejection). (College of LSA)
- **Mentor: 2** — Clear mentorship network (Bredig correspondence artifact; Rutherford year), but no single documented mentor “turning point” moment. (digital.sciencehistory.org)
- **Controversy: 3** — Strongly sourceable conflict: priority/naming complexities + late-career theory dismissed by academia but welcomed in industry. (Chemistry World)
- **Human Detail: 2** — Good documented character contrasts and teaching reputation, but limited open primary personal writing quoted so far. (College of LSA)

- **Last Experiment: 2** — Late-career seminars + sustained advocacy for quanticule theory are scene-able, though not a single "final act" document yet.

Primary recommended lens: Controversy

- **Why it fits:** Scene #3 (element-91 naming/credit complexity) + Scene #5 (quanticule theory reception split between academia and industry) give clear stakes and documented outcomes. (math.unt.edu)
- **Best opening scene: Scene #5** (industry seminars vs academic dismissal) or **Scene #3** (brevium/protactinium naming).
- **One quote to use early:** Quote #1 from his 1913 paper ("The starting point…") to show his drive to systematize messy phenomena. (ChemTeam)
- **Cautions:** Be careful not to overclaim "he discovered protactinium" without isotope nuance; and flag the 1935/1936 departure discrepancy. (math.unt.edu)

Backup lens (within 1 point): Eureka

- **Why it fits:** Scene #2 + quotes provide unusually direct access to the discovery logic in his own words. (ChemTeam)
- **Best opening scene: Scene #2** (writing the α/β framework). (ChemTeam)
- **One quote to use early:** Quote #2 ("divide all radioactive transformations into two classes…"). (ChemTeam)
- **Cautions:** Avoid inventing a dramatic "aha" moment—stick to what the text shows. (ChemTeam)

References (all sources used)

1. **Deep Research Prompt (course handout PDF)** — assignment constraints + required output format.
2. **Fajans, K. (1913). *Radioactive Transformations and the Periodic System of the Elements*** (translated excerpt from *Berichte der Deutschen Chemischen Gesellschaft*, vol. 46, pp. 422–439), reproduced at ChemTeam.
3. **University of Michigan, LSA Chemistry (Department History). "Kasimir J. Fajans (1887–1975)"** — education, Munich role, departure timing, teaching reputation, publication count.
4. **Science History Institute Digital Collections. "Postcard from Kasimir Fajans to Georg Bredig"** — archival artifact + metadata (includes migration context).
5. **Science History Institute Digital Collections. "Postcard of Kasimir Fajans and Bror Holmberg … shelves of glassware"** — visual primary artifact / lab texture.

6. **Bulletin for the History of Chemistry (1990), issue PDF** — quanticule theory reception, industry seminars, personality/mentorship notes, Pauling anecdote.
7. **Kragh, H. (2011). "The isotope effect: Prediction, discussion, and discovery" (arXiv preprint)** — notes Fajans spent 1910–1911 with Rutherford.
8. **Royal Society Publishing / Notes & Records (2020). Egdell, R. G. "Henry Moseley, X-ray spectroscopy and the periodic table"** — element-91 gap context; notes short-lived isotope identified by Fajans & Göhring (1913).
9. **Chemistry World. "Transmutations and isotopes"** — background/context on displacement law / isotope era.
10. **VanderKrogt Elements Multidict. "Protactinium" entry** — "brevium" naming rationale and summary of 1913 identification.
11. **Royal Society of Chemistry Periodic Table (RSC). "Protactinium"** — general element-history context (used for cross-checking the element-91 story).
12. **Encyclopaedia Britannica. "Fajans' rules"** — supporting background on the named rule (used as general reference context).
13. **General biographical cross-check (Britannica/other bio page used earlier for identity/context)** — Kasimir/Kazimierz name form + basic bio cross-check.

If you want, paste your draft paragraph(s) and I'll insert **inline** citations that match this end-reference list.